

Chapter 12 & 13: Structures & Mechanical Properties of Ceramics

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Materials Science (MIAE 221)

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapters 12 & 13

1. Executive Overview & First-Principles Philosophy

Ceramics are inorganic, non-metallic compounds formed primarily between metallic and non-metallic elements (oxides, nitrides, carbides, borides, and silicates).

From a chemical bonding perspective, ceramics are governed by **strong ionic bonds** (Coulombic attraction between cations and anions) and/or **highly directional covalent bonds**. This atomic architecture endows ceramics with engineering properties that metals and polymers cannot match:

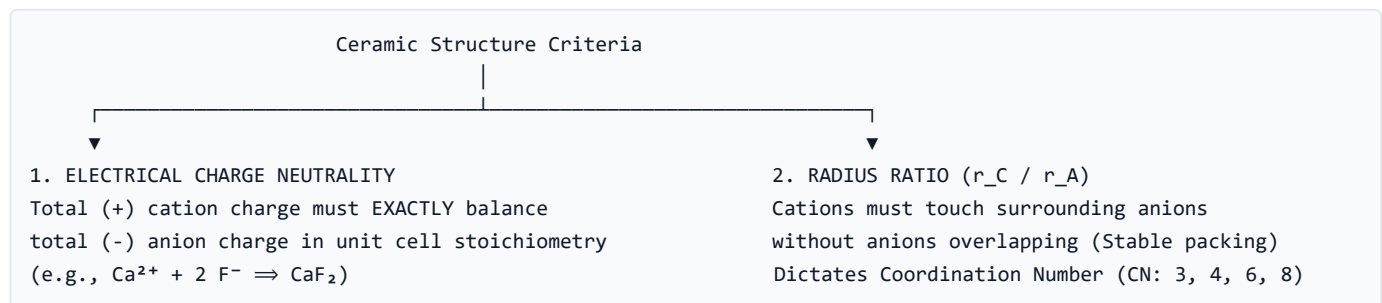
- Extreme hardness and wear resistance (e.g., Diamond, Silicon Carbide, Alumina).
- Exceptional thermal stability and ultra-high melting temperatures ($T_m > 2000^\circ\text{C}$).
- Chemical inertness and oxidation resistance in aggressive, corrosive environments.
- Electrical and thermal insulation.

However, these same atomic bonds create the primary engineering limitation of ceramics: **extreme brittleness and catastrophic sensitivity to microscopic flaws**. Because like-charge ions strongly repel one another, planes of ions cannot slide past each other without shattering the crystal lattice. Consequently, ceramics cannot be tested or designed using metallic tensile testing methods; they require specialized flexural testing and probabilistic fracture mechanics.

2. Ceramic Crystal Structures: The Geometric Packing Rules (Callister §12.2)

Unlike elemental metals where identical atoms pack into simple FCC or BCC unit cells, ceramic crystal structures consist of **at least two different chemical species**: positively charged metallic **cations** and negatively charged nonmetallic **anions**.

Two fundamental physical criteria govern whether a ceramic crystal structure is thermodynamically stable:



2.1 The Cation-to-Anion Radius Ratio (r_C/r_A)

Because metallic atoms shed valence electrons to become cations, **cations are significantly smaller than anions**:

$$r_C < r_A \implies \frac{r_C}{r_A} < 1.0$$

- **Stable Packing Condition:** For a crystal structure to be stable, the central cation must be in direct contact with all of its surrounding coordinating anions. If the cation is too small, the surrounding anions touch each other while the cation rattles loosely in the central void (**unstable packing**).
- The ratio of ionic radii $\frac{r_C}{r_A}$ dictates the geometric **Coordination Number (CN)** (the number of nearest-neighbor anions surrounding each cation):

Radius Ratio Range ($\frac{r_C}{r_A}$)	Coordination Number (CN)	Coordination Geometry	Archetype Crystal Structure
< 0.155	2	Linear	Rare
$0.155 - 0.225$	3	Planar Triangular	Boron Oxide (B_2O_3)
$0.225 - 0.414$	4	Tetrahedral	Zinc Blende (ZnS), Silicon Carbide (SiC)
$0.414 - 0.732$	6	Octahedral	Rock Salt ($NaCl$), Magnesia (MgO), FeO
$0.732 - 1.000$	8	Cubic	Cesium Chloride ($CsCl$)
$= 1.000$	12	Close-Packed (FCC/HCP)	Metals

2.2 Major Ceramic Crystal Structure Archetypes

A. Rock Salt ($NaCl$) Structure (AX -Type, $CN = 6$)

$$\overline{AO} = r_A + r_C$$

Furthermore, the side length ratio $\overline{AP}/\overline{AO}$ is a function of the angle α as

$$\frac{\overline{AP}}{\overline{AO}} = \cos \alpha$$

The magnitude of α is 30° , because line $\overline{AO}$ bisects the 60° angle BAC . Thus,

$$\frac{\overline{AP}}{\overline{AO}} = \frac{r_A}{r_A + r_C} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Solving for the cation–anion radius ratio,

$$\frac{r_C}{r_A} = \frac{1 - \sqrt{3}/2}{\sqrt{3}/2} = 0.155$$

AX-Type Crystal Structures

Some of the common ceramic materials are those in which there are equal numbers of cations and anions. These are often referred to as AX compounds, where A denotes the cation and X the anion. There are several different crystal structures for AX compounds; each is typically named after a common material that assumes the particular structure.

Figure 12.2: The Rock Salt ($NaCl$) unit cell. Anions (Cl^- , green) form an FCC lattice, while cations (Na^+ , purple) occupy all octahedral interstitial sites — from Callister & Rethwisch 10th Ed. (Fig. 12.2).

- **Geometry:** The larger anions form an **FCC unit cell array**. The smaller cations occupy the center of the unit cell and the centers of all 12 unit cell edges (the **octahedral interstitial sites**).
 - **Stoichiometry per Unit Cell:**
 - Anions: $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$ anions/cell.
 - Cations: $1 \text{ center} + 12 \times \frac{1}{4} = 4$ cations/cell.
 - Exactly 4 formula units (4 NaCl) per unit cell!
 - **Coordination Number:** CN = 6 for both cations and anions.
 - **Engineering Examples:** Sodium Chloride (NaCl), Magnesium Oxide (MgO refractory furnace bricks), Iron Oxide (FeO), Calcium Oxide (CaO), Titanium Carbide (TiC).
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B. Cesium Chloride (CsCl) Structure (AX-Type, CN = 8)

- **Geometry:** Large anions form a **Simple Cubic (SC)** array with a single large cation occupying the exact center of the cubic void.
 - **Stoichiometry:** 1 Cs⁺ and 1 Cl⁻ per unit cell.
 - **Coordination Number:** CN = 8 for both ions.
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C. Zinc Blende (ZnS) Structure (AX-Type, CN = 4)

- **Geometry:** Anions form an **FCC array**. Cations occupy **half of the tetrahedral interstitial sites** (4 of the 8 tetrahedral sites).
 - **Coordination Number:** CN = 4 (tetrahedral coordination, 109.5° bond angles).
 - **Engineering Examples:** ZnS, Silicon Carbide (SiC), Gallium Arsenide (GaAs), Indium Phosphide (InP).
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D. Fluorite (CaF₂) Structure (AX₂-Type, CN = 8 : 4)

- **Geometry:** Calcium cations (Ca²⁺) form an FCC array, while Fluoride anions (F⁻) occupy **all 8 tetrahedral interstitial sites**.
 - **Coordination Numbers:** Cation CN = 8 (surrounded by 8 F⁻); Anion CN = 4 (surrounded by 4 Ca²⁺).
 - **Engineering Examples:** CaF₂, Uranium Dioxide (UO₂ nuclear fuel pellets), Zirconia (ZrO₂ thermal barrier coatings).
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E. Perovskite (BaTiO₃) Structure (ABX₃-Type Ternary Oxide)

- **Geometry:** Barium (Ba²⁺) cations occupy the 8 cube corners, Oxygen (O²⁻) anions occupy the 6 face centers, and a small Titanium (Ti⁴⁺) cation sits at the body center.
 - **Engineering Significance:** Exhibits **piezoelectric and ferroelectric behavior**. Below 120°C (Curie temperature), the central Ti⁴⁺ ion shifts off-center, creating a permanent electric dipole used in sonar, ultrasonic transducers, and dielectric capacitors!
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3. Silicates & Carbon Allotropes (Callister §12.3 – §12.4)

3.1 Silicate Ceramics & Glass Networks

Silicates are the most abundant minerals in the Earth's crust (soils, clays, rocks, sands).

- **The Basic Building Unit: The Silicon-Oxygen Tetrahedron (SiO₄⁴⁻):**

- Because $r_{\text{Si}}/r_{\text{O}} \approx 0.29$ (lying in the range $0.225 - 0.414$), each Silicon ion is tetrahedrally bonded to **4 Oxygen ions**.
- Each oxygen ion carries a -1 unsatisfied charge, allowing tetrahedra to link by sharing corners.

oxides do not form polyhedral networks. Rather, their cations are incorporated within and modify the SiO_4^{4-} network; for this reason, these oxide additives are termed *network modifiers*. For example, Figure 12.11 is a schematic representation of the structure of a sodium–silicate glass. Still other oxides, such as TiO_2 and Al_2O_3 although not network formers, substitute for silicon and become part of and stabilize the network; these are called *intermediates*. From a practical perspective, the addition of these modifiers and intermediates lowers the melting point and viscosity of a glass and makes it easier to form at lower temperatures (Section 13.11).

The Silicates

For the various silicate minerals, one, two, or three of the corner oxygen atoms of the SiO_4^{4-} tetrahedra are shared by other tetrahedra to form some rather complex structures. Some of these, represented in Figure 12.12, have formulas SiO_4^{4-} , $\text{Si}_2\text{O}_7^{6-}$, $\text{Si}_3\text{O}_9^{6-}$ and so on; single-chain structures are also possible, as in Figure 12.12e. Positively charged cations such as Ca^{2+} , Mg^{2+} , and Al^{3+} serve two roles: First, they compensate the negative charges from the SiO_4^{4-} units so that charge neutrality is achieved; second, these cations ionically bond the SiO_4^{4-} tetrahedra together.

Figure 12.10: Two-dimensional schematic representation of (a) Crystalline silica (SiO_2) showing long-range periodicity, and (b) Noncrystalline (amorphous) silica glass showing random network disorder — from Callister & Rethwisch 10th Ed. (Fig. 12.10).

- **Silica (SiO_2) Allotropes:**

- In pure silica, **every single oxygen corner is shared between two tetrahedra**, yielding charge neutrality (SiO_2).
- *Crystalline Silica*

 - : Quartz, Cristobalite, Tridymite (ordered 3D frameworks).
- *Amorphous Silica Glass*

: When molten silica cools rapidly, the viscosity surges so quickly that tetrahedra cannot arrange into periodic lattices; they freeze into an amorphous, random network.

- **Network Formers vs. Network Modifiers in Industrial Glasses:**

- Pure silica glass (fused silica) has an extraordinarily high softening point ($> 1600^\circ\text{C}$) and extreme melt viscosity, making it expensive to shape.
- To produce commercial window glass (Soda-Lime Glass), **Network Modifiers** (Na_2O and CaO) are added:
 - The Na^+ and Ca^{2+} cations do not join the covalent tetrahedral network; they lodge in the interstitial gaps.
 - The extra O^{2-} ions break continuous $\text{Si} - \text{O} - \text{Si}$ bridging bonds, forming non-bridging terminal oxygens.
 - Breaking the network connectivity **slashes the processing temperature to $\sim 1000^\circ\text{C}$ and lowers melt viscosity**, enabling rapid, high-speed automated bottle and float glass manufacturing!

3.2 Carbon Allotropes: Diamond vs. Graphite vs. Graphene

Carbon exhibits allotropic polymorphism with the most extreme property divergences in the physical universe:

1. **Diamond:**

- All carbon atoms are covalently bonded in an sp^3 tetrahedral network (109.5° bond angles).

- Highest hardness of any known natural bulk material; extreme thermal conductivity ($k \approx 2000 \text{ W}/(\text{m} \cdot \text{K})$); wide electronic bandgap ($E_g = 5.5 \text{ eV}$, electrical insulator).

2. Graphite:

- Carbon atoms form flat, 2D hexagonal planar sheets bonded by strong sp^2 covalent bonds (120° angles).
- Adjacent parallel graphene sheets are held together only by **weak secondary van der Waals forces**.

◦ *Consequence*

: The sheets slide over each other with almost zero shear resistance, making graphite an exceptional solid dry lubricant. Delocalized π -electrons within sheets provide high electrical conductivity in-plane.

3. Graphene & Carbon Nanotubes:

◦ *Graphene*

: A single, isolated 2-dimensional monolayer of sp^2 -bonded carbon atoms. Exhibits theoretical tensile strength of $\approx 130 \text{ GPa}$ ($100\times$ stronger than steel!) and ballistic electron mobility.

◦ *Carbon Nanotubes (CNTs)*

: Graphene sheets rolled seamlessly into nanoscale cylinders.

4. Mechanical Properties of Ceramics: Flexural Testing & Flaw Theory (Callister §12.8)

4.1 Why Ceramics Cannot Be Tested in Direct Tension

Conducting standard uniaxial tensile tests on ceramic specimens is virtually impossible in engineering laboratories for three reasons:

1. Gripping a brittle ceramic in mechanical test jaws induces localized contact stresses that crush the ends of the specimen before the test begins.
2. Even a microscopic misalignment ($< 0.1^\circ$) introduces bending moments that cause premature tensile fracture on the outer surface.
3. Ceramics cannot be easily machined into round dogbone specimens without introducing surface microcracks.

4.2 The Three-Point Bending Test (Flexural Strength)

To overcome these limitations, the mechanical strength of ceramics is measured using a **Three-Point Bending Test** (also called **Modulus of Rupture** or **Transverse Rupture Strength**):

raised, the magnitude of the bonding is diminished, the sliding motion or flow of the atoms or ions is facilitated, and subsequently there is an attendant decrease in viscosity. A discussion of the temperature dependence of viscosity for glasses is deferred to Section 13.11.

Figure 12.32 Representation of the viscous flow of a liquid or fluid glass in response to an applied shear force.

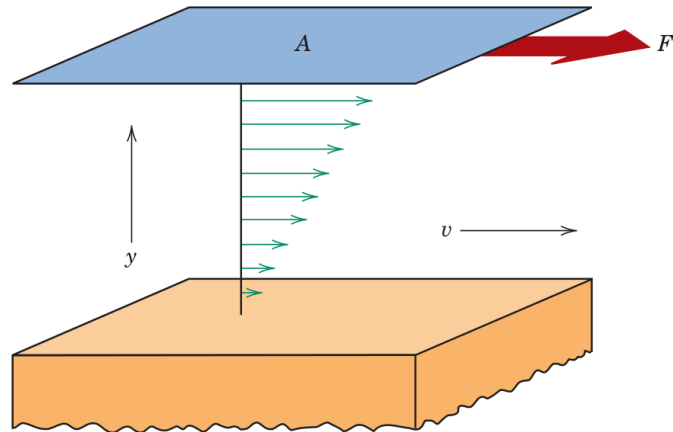


Figure 12.32: Schematic diagram of a Three-Point Bending Test on a ceramic beam supported at span distance L and loaded at the midpoint with fracture force F_f — from Callister & Rethwisch 10th Ed. (Fig. 12.32).

A rectangular or circular beam resting on two support pins spaced at span length L is loaded at its midpoint by an applied vertical force F :

- The top surface of the beam is under **compression**.
- The bottom surface is under **tension**, where crack initiation and brittle fracture occur.

Flexural Strength (σ_{fs}) Formulas:

1. For a Beam with Rectangular Cross-Section (Width b , Depth d):

$$\sigma_{fs} = \frac{3F_f L}{2bd^2}$$

2. For a Beam with Circular Cross-Section of Radius R :

$$\sigma_{fs} = \frac{F_f L}{\pi R^3}$$

where:

- σ_{fs} = Flexural Strength / Modulus of Rupture (MPa).
- F_f = load at catastrophic fracture (N).
- L = support span length (mm).
- b, d, R = specimen cross-sectional dimensions (mm).

4.3 Influence of Porosity on Mechanical Properties

During ceramic manufacturing, powders are pressed and sintered. Unless hot isostatic pressing is used, sintered ceramics inevitably contain residual microscopic **pores (voids)**.

Porosity has a catastrophic degradation effect on both stiffness and strength:

Porosity Effects

1. ELASTIC MODULUS REDUCTION

Pores carry zero elastic load

$$E = E_0 (1 - 1.9 P + 0.9 P^2)$$

10% porosity cuts E by ~20%

2. FLEXURAL STRENGTH REDUCTION

Pores act as severe stress concentrators

$$\sigma_{fs} = \sigma_0 \exp(-n P) \quad (n \approx 4 - 7)$$

10% porosity cuts strength by ~50%!

1. Reduction in Elastic Modulus (E):

$$E = E_0 (1 - 1.9P + 0.9P^2)$$

where E_0 is the modulus of the 100 dense, pore-free ceramic, and P is the volume fraction of porosity ($0 \leq P \leq 1$).

2. Exponential Collapse in Flexural Strength (σ_{fs}):

Pores not only reduce the load-bearing cross-sectional area; they act as **sharp internal stress concentrators (Griffith crack flaws)** from which brittle fracture nucleates:

$$\sigma_{fs} = \sigma_0 \exp(-nP)$$

where σ_0 is the strength of the fully dense ceramic, and n is an empirical constant typically between 4 and 7.

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Dramatic Consequence

: Because of the exponential factor $\exp(-nP)$, **introducing merely 10 porosity ($P = 0.10$) slashes the flexural strength of alumina by more than 50!**

5. Comprehensive Step-by-Step Problem Walkthroughs

5.1 Problem 1: Predicting Ceramic Crystal Structure from Ionic Radii

Problem Statement: Magnesium Oxide (MgO) and Iron Oxide (FeO) are important industrial ceramic refractories. Given the ionic radii:

- $r_{\text{Mg}^{2+}} = 0.072 \text{ nm}$
- $r_{\text{Fe}^{2+}} = 0.077 \text{ nm}$
- $r_{\text{O}^{2-}} = 0.140 \text{ nm}$

1. Calculate the cation-to-anion radius ratio for both oxides.
2. Predict the coordination number (CN) for each oxide.
3. Identify the expected crystal structure.
4. Calculate the theoretical density of MgO, given atomic weights $A_{\text{Mg}} = 24.31 \text{ g/mol}$ and $A_{\text{O}} = 16.00 \text{ g/mol}$.

Step 1: Calculate the Radius Ratios

- For MgO:

$$\frac{r_{\text{Mg}^{2+}}}{r_{\text{O}^{2-}}} = \frac{0.072 \text{ nm}}{0.140 \text{ nm}} = 0.514$$

- For FeO:

$$\frac{r_{\text{Fe}^{2+}}}{r_{\text{O}^{2-}}} = \frac{0.077 \text{ nm}}{0.140 \text{ nm}} = 0.550$$

Step 2: Determine Coordination Number (CN)

Both ratios (0.514 and 0.550) fall within the octahedral range:

$$0.414 \leq \frac{r_C}{r_A} < 0.732$$

Therefore, both oxides have a **Coordination Number of 6** (CN = 6) for both cations and anions.

Step 3: Identify the Crystal Structure

Because the stoichiometry is 1 : 1 (AX-type) and CN = 6, both MgO and FeO crystallize into the **Rock Salt** (NaCl) **crystal structure**!

Step 4: Calculate the Theoretical Density of MgO

In the Rock Salt unit cell, cations and anions touch along the cube edge:

$$a = 2(r_C + r_A) = 2(0.072 \text{ nm} + 0.140 \text{ nm}) = 2(0.212 \text{ nm}) = 0.424 \text{ nm} = 4.24 \times 10^{-8} \text{ cm}$$

Unit cell volume:

$$V_c = a^3 = (4.24 \times 10^{-8} \text{ cm})^3 = 7.623 \times 10^{-23} \text{ cm}^3$$

There are $n = 4$ formula units of MgO per unit cell:

$$M_{\text{cell}} = 4 \times (A_{\text{Mg}} + A_{\text{O}}) = 4 \times (24.31 + 16.00) = 4 \times 40.31 = 161.24 \text{ g/mol}$$

$$\rho = \frac{n(A_{\text{Mg}} + A_{\text{O}})}{V_c N_A} = \frac{161.24 \text{ g/mol}}{(7.623 \times 10^{-23} \text{ cm}^3) \times (6.022 \times 10^{23} \text{ atoms/mol})} = \frac{161.24}{45.906} = 3.512 \text{ g/cm}^3 \approx 3.51 \text{ g/cm}^3$$

5.2 Problem 2: Three-Point Bending Test & Porosity Degradation

Problem Statement: A three-point bending test is performed on an alumina (Al_2O_3) ceramic specimen having a rectangular cross-section of width $b = 10.0 \text{ mm}$ and depth $d = 5.0 \text{ mm}$, supported across a span $L = 50.0 \text{ mm}$.

1. If the specimen fractures at an applied center load $F_f = 400 \text{ N}$, compute the flexural strength (σ_{fs}) of this ceramic.
2. If this tested specimen has a porosity volume fraction $P = 0.08$ (8 porosity) and empirical exponent $n = 5.0$, determine the theoretical flexural strength (σ_0) of a completely dense (0 porosity) alumina specimen.
3. Predict the flexural strength if the porosity rises to $P = 0.15$ (15).

Step 1: Compute Flexural Strength (σ_{fs})

Using the rectangular three-point bending formula:

$$\sigma_{fs} = \frac{3F_f L}{2bd^2}$$

Substitute $F_f = 400 \text{ N}$, $L = 50.0 \text{ mm}$, $b = 10.0 \text{ mm}$, $d = 5.0 \text{ mm}$:

$$d^2 = (5.0 \text{ mm})^2 = 25.0 \text{ mm}^2$$

$$\sigma_{fs} = \frac{3 \times (400 \text{ N}) \times (50.0 \text{ mm})}{2 \times (10.0 \text{ mm}) \times (25.0 \text{ mm}^2)} = \frac{60,000}{500} = 120.0 \text{ N/mm}^2 = 120 \text{ MPa}$$

Step 2: Determine Fully Dense Strength (σ_0)

Using the porosity exponential equation:

$$\sigma_{fs} = \sigma_0 \exp(-nP) \implies \sigma_0 = \frac{\sigma_{fs}}{\exp(-nP)} = \sigma_{fs} \exp(nP)$$

Substitute $\sigma_{fs} = 120$ MPa, $n = 5.0$, $P = 0.08$:

$$nP = 5.0 \times 0.08 = 0.40$$

$$\sigma_0 = 120 \times \exp(0.40) = 120 \times 1.4918 = 179.0 \text{ MPa}$$

Result

: A fully dense, pore-free alumina specimen exhibits a flexural strength of 179 MPa.

Step 3: Predict Flexural Strength at $P = 0.15$

$$nP_2 = 5.0 \times 0.15 = 0.75$$

$$\sigma_{fs}(0.15) = \sigma_0 \exp(-0.75) = 179.0 \times 0.4724 = 84.55 \text{ MPa} \approx 84.6 \text{ MPa}$$

Result

: Increasing porosity from 8 to 15 slashes the flexural strength from 120 MPa down to 84.6 MPa!

6. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Three-Point Bending: Depth Squared vs. Width Squared:** In $\sigma_{fs} = \frac{3FL}{2bd^2}$, the **vertical beam depth d is squared**, NOT the horizontal width b ! Swapping b and d will produce a catastrophic mathematical error!
- **⚠ Trap 2: Ionic Touch Geometry in Density Calculations:** In Rock Salt (NaCl and MgO), cations and anions touch along the cube edge:

$$a = 2(r_C + r_A)$$

Do NOT use metallic formulas like $a = 2R\sqrt{2}$ or $a = 4R/\sqrt{3}$! Those apply only to single-element metals.

- **⚠ Trap 3: Porosity Fraction vs. Percentage:** In $\sigma_{fs} = \sigma_0 \exp(-nP)$, P is the **volume fraction**, NOT the percentage! If an exam states 8 porosity, substitute $P = 0.08$. Substituting $P = 8$ will compute $\exp(-40) \approx 0$, yielding zero strength!
- **⚠ Trap 4: Circular vs. Rectangular Bending Formulas:** Check specimen geometry carefully!
 - Rectangular: $\sigma_{fs} = \frac{3FL}{2bd^2}$.
 - Cylindrical rod: $\sigma_{fs} = \frac{FL}{\pi R^3}$ (where R is radius, NOT diameter).